Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/04/2006
PI Name:	Ahlers, Guenter Ph.D.		
Project Title:	The Superfluid Transition of 4He under Unusual Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2001	**End Date:**	03/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2903		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We continued our refinement of methods of heat capacity measurements of helium confined in cylindrical tubes. Measurements have been made for tubes of 1 micro-m diameter. These results are required for the theory of the fnite-size effects on the hermal conductivity which we also study with support from NASA. Plans are under way also for similar measurements for helium confined in cylindrical channels of 5 micro-m cross section. Having measurements for two different sizes will enable us to learn whether finite-size scaling is applicable to this system. The objective of this project is to study the superfluid transition in finite geometries. The goal is to determine the effect of confinement of the sample in micron-sized geometries of different dimensionality on the transition. A comparison with ongoing theoretical calculations will shed light on the boundary conditions which pertain to the fluid-solid interface. A greater understanding of the effect of boundaries on properties of liquid helium is relevant to the experiments BEST and DYNAMX. In		

	addition, our measurements are important to fundamental science in that they provide data to test theoretical predictions based on the renormalization-group theory. They are particularly important in testing the validity of the boundary conditions between a solid and the fluid that are used in the theory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A comparison between experimental results and theoretical prdictions will enhance our ability to calculate properties of complex systems using renormalization-group and Monte Carlo methods.
Task Progress:	We finalized the development of calorimetric methods for the determination of small heat capacities with high precision. We completed measurements of the heat capacity of liquid He-4 contained in the 1 micro-m diameter cylindrical capillaries of a glass capillary array. Data analysis is near its final stages. The results are largely consistent with previous measurements by Lipa and co-workers. That work had suggested a departure from finite-size scaling for the heat capacity near the superfluid transition in finite geometries. We modified our apparatus and installed a new glass capillary array with cylindrical pores of 5 micro-m diameter. Measurements for this geometry will add to our understanding and knowledge of departures from finite-size scaling in this system.
Bibliography Type:	Description: (Last Updated: 02/06/2013)